

CryptoCast: GUVI Live Evaluation Guide

GUVI × HCL Master Data Science Program — Complete Capstone Defense Blueprint

1. GUVI Live Evaluation & Scoring Matrix (Total: 60 Marks)

Process Step	Duration	Key Strategy / Focus Area
1. Self Introduction	3 Mins	State name, background, domain interest (Quant Finance / Deep Learning).
2. Project Problem Statement & Explanation	12 Mins	Follow the mandatory GUVI 13-step model building sequence.
3. Mock Viva Questions	10 Mins	1-2 deep technical questions per technology (Python, PyTorch, SciPy, Streamlit).
4. Feedback & Evaluation Scoring	5 Mins	Review scoring notes and ZEN portal entry confirmation.

Sl.	Evaluation Metric	Marks	How CryptoCast Achieves Top Marks
1	Code Quality / Data Transformations	10	Leak-free scaling, stationary log returns, PEP-8 modular functions.
2	Proper Documentation (Git ReadMe/PPT)	10	100% committed ReadMe with 11 dashboard pages & multi-horizon tables.
3	Code Reusability (Modular Programming)	10	Decoupled `src/` directory with reusable strategy & model classes.
4	Presentation	10	Interactive 11-page Streamlit web app with custom glassmorphism dark theme.
5	Task Accomplishment	10	All 5 DL models trained & evaluated across 1D, 3D, and 7D horizons.
6	5 Mock Questions (Viva)	10	Clear answers for loss functions, RevIN, R^2 paradox, and stationarity.
TOTAL SCORE		60 Marks	Goal: 60 / 60

2. Mandatory GUVI 13-Step Model Building Speech (12-Minute Script)

Follow this exact step-by-step sequence during your 12-minute project explanation:

1. Domain Intro (3 Lines)

Cryptocurrency markets represent a highly volatile, 24/7 decentralized asset class. Price discovery is driven by micro-structural momentum, macro liquidity cycles, and halving supply shocks. Applying quantitative data science to crypto requires overcoming non-stationarity and extreme return tail-risk.

2. Problem Statement (1 Line)

To design, train, and compare multi-output deep learning models that predict Bitcoin closing prices across 1-day, 3-day, and 7-day horizons simultaneously.

3. Data Cleaning & Preprocessing

Cleaned ~5,000 daily records (2010–2024). Converted raw string prices with commas to float, parsed volume suffixes (K, M, B), and enforced chronological 80/20 train-test splitting. Crucially, MinMaxScaler was fit ONLY on training data to strictly prevent data leakage.

4. EDA Findings (Univariate/Bivariate/Multivariate)

Prices are heavily right-skewed and non-stationary. Daily returns show high leptokurtosis (heavy tails). Seasonality analysis revealed a strong Turn-of-the-Month (TOM) liquidity effect with Q4 (days 23-31) producing highest average daily returns (+0.775%).

5. Feature Engineering

Engineered 3 deterministic whitepaper protocol features derived from Satoshi Nakamoto's code: `Block\_Reward` (50->25->12.5->6.25->3.125), `Days\_Since\_Halving`, and `Halving\_Progress` (0.0 to 1.0). Unlike reactive technical indicators (RSI/SMA), these are known ahead of time.

6. Statistical Significance & Tests

Ran parametric (Student's t-test, Welch's t-test, 1-Way ANOVA) and non-parametric (Mann-Whitney U) hypothesis tests across halving epochs. Confirmed statistically significant return differences across halving regimes ($p < 0.05$).

7. Target & Class Formulation

Transformed raw prices into stationary log returns: $r = \ln(P[t+h] / P[t])$. Created long/short directional targets for trading signal evaluation and directional accuracy score calculation.

8. Base Model Selection

Chose 1D-CNN (3 Conv1D blocks with causal padding) as the initial benchmark model due to its rapid training speed and local pattern extraction capability.

9. Deep Learning Models Implemented

Implemented 5 PyTorch architectures: 1D-CNN, Simple RNN, LSTM (gated memory highway), Transformer (multi-head self-attention), and PatchTST (patching + Reversible Instance Normalization RevIN). Used Adam optimizer, ReduceLROnPlateau, and gradient clipping ($\text{max_norm}=1.0$).

10. Model Evaluation Metrics & Custom Loss

Evaluated MAE, RMSE, and MAPE. Designed a custom `DirectionalMSELoss` combining MSE with a directional penalty ($\alpha=0.15$) to penalize wrong trade direction predictions more heavily than magnitude errors.

11. Final Model Selection & Rationale

PatchTST is selected as the champion model. It beats all models at 3D (MAPE 3.85%) and 7D (MAPE 5.93%) horizons while matching LSTM at 1D (MAPE 2.06%). RevIN eliminates distribution shift, and patching reduces token attention complexity by 96% ($O(60^2) \rightarrow O(5^2)$).

12. Conclusion & Feature Importance

SHAP feature importance analysis confirms that `Block\_Reward` and `Price` drive long-term structural forecasts. Halving-aligned Walk-Forward Validation confirmed model stability across unseen market regimes.

13. Business Suggestion & Solution

Deployed a 3-state (Long/Short/Cash) Trading Simulator with dynamic position sizing on Page 10. Outperformed passive Buy-and-Hold by avoiding bear market drawdowns, providing actionable ROI for crypto hedge funds.

3. GUVI 15 EDA Questions Cheat Sheet

- Q: 1. Basic Characteristics?** — 4,964 daily observations across 9 feature columns (Aug 2010 to Mar 2024).
- Q: 2. Overall Structure?** — Time-series index with continuous floats for price/volume and discrete protocol values.
- Q: 3. Patterns in Data?** — Exponential price growth interrupted by 4-year post-halving bull-bear cycles.
- Q: 4. Presence of Outliers?** — Extreme daily price shocks (e.g. +25% or -30% single-day moves) present in return distribution.
- Q: 5. Missing Values?** — Zero missing values in primary BTC dataset after cleaning raw Investing.com formatting.
- Q: 6. Correctness of Data?** — Verified OHLC relationships ($\text{High} \geq \text{Max}(\text{Open}, \text{Close})$, $\text{Low} \leq \text{Min}(\text{Open}, \text{Close})$).
- Q: 7. Variable Correlations?** — Open, High, Low, and Close correlate near 0.99 with raw price; returns exhibit near zero autocorrelation.
- Q: 8. Comparison to Past Performance?** — Post-2020 institutional era shows lower daily return variance compared to early 2011-2013 era.
- Q: 9. Seasonality Present?** — Turn-of-the-Month (TOM) effect identified: Q4 (Days 23-31) and Q1 (Days 1-7) show higher returns.
- Q: 10. Feature Variability?** — Daily volume varies across 6 orders of magnitude (\$10K in 2010 to \$50B+ in 2024).
- Q: 11. Discrepancies Observed?** — Raw prices fail stationarity tests (ADF test $p > 0.05$); log returns pass (ADF $p < 0.001$).
- Q: 12. Unexpected Results?** — Simple RNN outperforms Transformer at 7D horizon due to Transformer overfitting on small sequence samples.
- Q: 13. Subset Behaviors?** — Pre-halving vs. post-halving subsets show statistically significant return variance differences.
- Q: 14. Required Transformations?** — Log return transformation $r = \ln(P[t+h]/P[t])$ and MinMaxScaler to range [0, 1].
- Q: 15. Gaps Identified?** — Lack of order book depth data; mitigated by incorporating macroeconomic liquidity proxies.

4. Technology Mock Viva Questions & Answers

Q1: Explain DirectionalMSELoss formula and why it's critical.

$\text{Loss} = \text{MSE} + \alpha * \text{mean}(\text{ReLU}(-y_{\text{pred}} * \text{sign}(y_{\text{true}})))$. Standard MSE only measures magnitude error. In trading, predicting 'up' when market drops 'down' causes capital loss. The ReLU term adds extra penalty whenever signs disagree ($\alpha=0.15$).

Q2: What is the R^2 Paradox in financial time series?

Evaluating predicted vs actual raw prices gives $R^2 \approx 0.99$ because yesterday's price is highly predictive of today's price level (random walk baseline). Evaluating predicted vs actual daily returns gives $R^2 \approx 1\text{-}3\%$. In quantitative finance, explaining 2% of daily return variance is considered highly successful.

Q3: How does PatchTST solve the attention bottleneck?

Standard Transformers treat each daily timestep as a token (60 tokens $\rightarrow O(60^2) = 3600$ attention operations). PatchTST groups 12 days into 1 patch (5 patches $\rightarrow O(5^2) = 25$ operations), reducing complexity by 96% while capturing local semantic sub-series.

Q4: Explain Reversible Instance Normalization (RevIN).

RevIN subtracts mean and divides by std of each input instance before feeding it to the network, and adds mean/std back after prediction. This handles non-stationary distribution shifts between training and live market regimes.

Q5: What is Walk-Forward Validation with Halving-Aligned Folds?

Standard k-fold cross-validation breaks time order (causing data leakage). WFV uses an expanding window that trains on past data and tests on future data. Halving-aligned folds ensure each test window covers an entire 4-year bull-bear halving epoch.